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RESEARCH ARTICLE

Co-designing, developing, and implementing multiple learning analytics dashboards for data-driven decision-making in education: a design-based research approach

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Abstract
This research investigates the design, development and implementation of Multiple Learning Analytics Dashboards (MLADs) with the goal of enhancing data-driven decision-making among teachers in primary education. The study presents a Design-Based Research (DBR) approach to ensure the iterative development and refinement of MLADs through collaboration with educational professionals. The research involved four key steps in the presented DBR approach: 1) identifying the requirements through stakeholder interviews; 2) generating design ideas through brainstorming sessions and prototyping; 3) collaboratively designing and developing MLADs and conducting usability testing to gather feedback; and 4) implementing the MLADs in real-world educational settings. This paper outlines the research methodology, the participants involved, and the progress made toward developing these Learning Analytics Dashboards (LADs). The outcomes highlight the creation of tailored MLADs for teachers across multiple municipalities, as well as the lessons learned from real-world implementation. Additionally, we analyze user feedback from teachers regarding the dashboard's clarity, navigation, functionality, and design, providing a comprehensive view of the dashboard's usability and areas for future enhancement.

Keywords Design-based research · Learning analytics dashboards · Primary education · Co-DESIGN · User-centered design · Data visualization · Learning analytics · EasyVis

Introduction

In recent years, Learning Analytics Dashboards (LADs) have emerged as powerful tools in education. They can provide actionable insights into student learning behaviors, performance, and progress (Viberg & Grönlund, 2021). In fact, by leveraging the increasing

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availability of digital data generated from Digital Learning Materials (DLMs) in educational settings, LADs can support teachers in making data-informed pedagogical decisions that enable more personalized learning experiences and targeted interventions (Camacho et al., 2020; Feldman, 2023; Ifenthaler & Yau, 2020; Mohseni et al., 2024).

While much of the research on Learning Analytics (LAs) has focused on higher education, there is a growing interest in adapting LADs for primary and secondary education (Karademir et al., 2024; Mohseni et al., 2024). However, research in this educational context remains limited, with existing dashboards often failing to address the specific needs of younger students and their teachers (Karademir et al., 2024; Mandinach & Gummer, 2016; Viberg & Grönlund, 2021). For primary school teachers, who are often pressed for time, having access to actionable data can be instrumental in identifying learning difficulties, providing timely feedback, and supporting differentiated instruction (Ata & Alpaslan, 2024).

Despite the potential benefits of LADs, teachers frequently encounter substantial barriers to effective data-driven decision-making, such as limited time, inadequate data literacy, and insufficient training in data interpretation (Mandinach & Gummer, 2016). These challenges are particularly acute in primary education settings, where diverse student learning needs and intensive instructional demands further limit teachers' ability to analyze and interpret complex educational data effectively and align it with their pedagogical practices (Van Leeuwen, 2019). Consequently, there is an urgent need for tools that not only simplify data visualization but also closely align with curricular objectives and instructional strategies specific to primary education contexts (Masiello et al., 2024). In response to these challenges, this study aims to co-design and develop Multiple Learning Analytics Dashboards (MLADs), to streamline data access and visualization and to provide pedagogically actionable insights within teachers' constrained time frames. The term MLADs refers to having different LADs tailored to the needs of teachers in three municipalities in Sweden, with each LAD containing multiple pages. The reason for selecting these municipalities was their differences in digital infrastructure, DLM platforms, and student demographics, which provided diverse contexts to test the adaptability and scalability of the dashboards. Recent research underscores that dashboards developed through a teacher-centered design process significantly alleviate data fatigue and foster greater adoption of analytics at the classroom level (Ahn et al., 2019; Lee-Cultura et al., 2023; Mohseni et al., 2023).

Although the aim was to create LADs (referred to as EasyVis dashboards) as consistent as possible, differences in the data and DLMs across schools required some variations in the visualizations and features. Rather than being a limitation, these variations highlight the methodological importance of adaptable and modular dashboard design within a DBR approach (McKenney & Reeves, 2018). By tailoring visualizations to local infrastructures and pedagogical practices, the MLADs were able to address the unique contexts and needs of each school while still following a shared design framework. This adaptability ensures that the dashboards provide effective support for teaching and learning environments, while also demonstrating how DBR can balance local customization with scalability. While the initial implementation is context-specific, the approach and findings offer potential for broader application beyond the local setting. These dashboards integrate data visualization with evidence-based educational theories, enabling teachers to gain insights that are directly actionable in their classrooms (Gašević et al., 2015).

The study follows a Design-Based Research (DBR) methodology, which is particularly well-suited for the iterative development of educational technology (Mohseni, 2024; Wang & Hannafin, 2005). DBR emphasizes a collaborative process involving researchers, teachers, and other stakeholders to ensure that the solutions developed are grounded in

real-world contexts and responsive to teachers' needs (Amiel & Reeves, 2008; Anderson & Shattuck, 2012). This approach allows for continuous refinement of the dashboards through cycles of design, implementation, and evaluation, ensuring that the final product is not only theoretically sound but also practically relevant and scalable for everyday classroom use (McKenney & Reeves, 2018; Wang & Hannafin, 2005). By using a collaborative design approach grounded in DBR, this research strives to bridge the gap between theory and practice in the use of LAs in primary education. The goal is to create EasyVis dashboards that not only provide valuable insights for teachers but also contribute to the growing body of knowledge on the effective use of LAs in K-12 settings, ultimately supporting improved educational outcomes for students (Sandoval, 2014).

This study makes the following key contributions: (1) A co-design methodology that actively engaged teachers in collaborative activities such as ideation, prototyping, and iterative feedback, ensuring that MLADs were grounded in teachers' pedagogical needs; (2) An iterative, user-centered DBR framework that structured these co-design activities into cycles of design, implementation, and evaluation involving teachers, school leaders, and EdTech companies; (3) The implementation and evaluation of MLADs tailored to three different municipalities with varying data and pedagogical contexts; and (4) Evidence on the usability, scalability, and contextual relevance of MLADs for data-informed decision-making in education.

The rest of the article is organized as follows. Sects. "Related work" and "Methodology" describe the related work and the research methodology, respectively. In Sects. "Outcomes" and "Discussion" we account for the outcomes and discussion, respectively. Sect. "Conclusion and future work" concludes the paper and discuss the future work.

Related work

The iterative nature of DBR blends theory and practice to produce practical solutions while advancing theoretical understanding (Anderson & Shattuck, 2012; Campanella & Penuel, 2021; Wang & Hannafin, 2005). DBR involves cycles of design, implementation, evaluation, and refinement, making it well-suited for tackling complex, real-world educational challenges (Collins et al., 2016; Mohseni, 2024). It has been effectively used to develop technology-enhanced learning tools, classroom interventions, and pedagogical frameworks tailored to the needs of teachers and students (McKenney & Reeves, 2018; Reinking, 2021).

One significant body of research on DBR focuses on its use for integrating DLMs into educational environments to enhance student engagement and learning outcomes (Wang, 2020). For example, Hoadley (2004) explored how DBR could be used to design collaborative learning technologies in secondary education settings. The study emphasized the importance of involving teachers and students in the design process to ensure that the resulting tools are user-centered and practical. However, challenges related to scalability and teacher training were noted, which limited the broader adoption of the interventions developed. This aligns with findings from subsequent studies that highlight the need for professional development as a critical factor in the successful implementation of technology in education (Akram et al., 2022).

Despite the successes, several challenges have been identified in using DBR in primary and secondary education settings. One common issue is the difficulty of sustaining teacher engagement throughout the DBR process, especially when projects extend over multiple

academic years (Design-Based Research Collective, 2003). The need for sustained collaboration between researchers and practitioners is critical, as highlighted by Barab and Squire (2016), but it is often challenging to maintain due to time constraints and the varying priorities of teachers and researchers. Furthermore, many DBR studies have faced challenges in balancing the rigor of scientific research with the practical demands of classroom implementation (Brown, 1992). Ensuring that the findings are both theoretically robust and practically useful remains a key tension in the field.

In addition to technology integration, DBR has been applied to develop new pedagogical strategies in secondary education. For instance, Anderson and Shattuck (2012) used DBR to create inquiry-based science learning environments that encourage critical thinking and student autonomy. The iterative nature of DBR allowed for continuous feedback and adaptation, which was critical in refining the instructional strategies (Jacobs et al., 2024). However, a common challenge identified was the time-intensive nature of the iterative cycles, which can be difficult to manage within the constraints of a typical school schedule (Anderson & Shattuck, 2012).

Moreover, DBR approaches have been used to address the need for inclusivity and differentiation in primary classrooms. For example, Sandoval and Bell (2004) applied DBR to design science curricula that accommodate diverse student needs. The study demonstrated that using DBR enabled the development of adaptable curricular materials that could be tailored to different student groups. However, the researchers noted that achieving this level of customization required significant resources and teacher training, which may not always be feasible in under-resourced schools (Ford et al., 2017; Gynther, 2016; Masiello et al., 2023).

As previously mentioned, the application of DBR in primary and secondary education has demonstrated significant potential in creating interventions that are both theoretically sound and practically useful. However, several challenges remain, particularly related to scalability, teacher engagement, and the resource-intensive nature of iterative design processes (Reinking, 2021). This study emphasizes the co-design and development of MLADs, highlighting the collaboration between researchers and teachers through a DBR approach.

Co-design has been increasingly emphasized in educational research as a way to ensure that learning technologies are relevant, usable, and adopted in practice. Prior studies have shown that co-design processes allow teachers to take ownership of tools, leading to improved usability and alignment with classroom needs (Holstein et al., 2019; Sanders & Stappers, 2008). At the same time, the literature highlights persistent challenges. For example, Nicholson et al. (2022) observed that co-design requires negotiating diverse priorities between researchers and practitioners, while Robertson and Simonsen (2012) noted that limited time and differing technical expertise can hinder sustained teacher participation. Despite these challenges, co-design remains a powerful strategy for bridging the gap between theoretical models and classroom realities (Smeenk, 2023). Our study builds on this evidence base by embedding co-design into all stages of the DBR process, while extending prior work through a multi-municipality approach that engaged teachers, pedagogical experts, IT staff, and EdTech partners in developing context-specific MLADs.

In primary education, teachers often lack timely, actionable insights into student engagement and progress, which limits their ability to intervene effectively (Ata & Alpaslan, 2024; Mandinach & Gummer, 2016). LADs have the potential to support formative assessment and differentiated instruction by visualizing patterns that are otherwise hard to detect. Differentiated instruction emphasizes adapting teaching strategies, content, and pacing to the diverse needs of students within the same classroom (Tomlinson, 2014). The LADs

operationalize this framework by offering student-level predictive alerts and class-average benchmarks, which enable teachers to identify struggling students early and adapt their instruction accordingly. Similarly, formative assessment refers to the ongoing collection and use of evidence of student learning to provide feedback that informs next instructional steps (Black & Wiliam, 2009; Wiliam, 2011). The LADs enhance this process by visualizing real-time trends in student performance, attendance, and engagement, thereby allowing teachers to provide timely, targeted feedback. However, many LADs assume a high level of data literacy (van Leeuwen, 2019), creating a gap between potential and practice. Therefore, involving teachers in the design of LADs is essential—not only to ensure usability but also to build trust and promote adoption (Holstein et al., 2019; Sanders & Stappers, 2008).

Previous studies in higher education offer valuable lessons on the visualization of performance and engagement (Ifenthaler & Schumacher, 2016; Viberg & Grönlund, 2021), which we adapt for primary settings in several ways. First, we simplified graphs and reduced the number of displayed indicators to avoid information overload, a challenge often reported by teachers with limited data literacy (Kippers et al., 2018). Second, we aligned dashboard metrics more closely with curriculum goals and assessment practices to ensure pedagogical relevance in everyday classroom planning (Ifenthaler & Yau, 2020). Third, we emphasized clarity of navigation and reduced cognitive load through streamlined dashboard layouts, reflecting principles highlighted in usability research on educational technologies (Bodily et al., 2019). Finally, we integrated real-time alerts and predictive insights to support the fast-paced feedback cycles required in primary classrooms, which is critical given teachers' time constraints and the importance of timely formative interventions (Datnow & Hubbard, 2016). Importantly, these adaptations were not only research-driven but also the result of iterative co-design sessions with teachers, who directly contributed to decisions about which metrics to prioritize, how visualizations should be simplified, and how the dashboards should align with curriculum and everyday practice. This research builds on that foundation by tailoring LADs for teachers of younger students with limited time and varying levels of digital fluency. On the other hand, some studies (Karademir et al., 2024; Viberg & Grönlund, 2021) emphasize the limitations of existing dashboards like Google Classroom Analytics and Edmodo Dashboard. While Google Classroom Analytics provides teachers with basic statistics on student grades and assignment completion, its main benefit lies in seamless integration with classroom management tools. However, it offers little real-time insight into student learning processes and only limited support for pedagogical alignment (Viberg & Grönlund, 2021). Similarly, Edmodo Dashboard includes features such as tracking logins, posts, and assignment submissions, and its strength has been in creating an accessible, socially oriented learning environment. Nevertheless, its analytics functions remain relatively surface-level, focusing on activity counts rather than supporting instructional decision-making or differentiated teaching (Karademir et al., 2024). Recent studies in primary education further underscore the relevance of these concerns. Paavilainen et al. (2024) demonstrated how a learning design perspective can align dashboard indicators with pedagogical goals to scaffold self-regulated learning (SRL) in primary classrooms. Dülger et al. (2025) extended this line of work by designing a classroom-level teacher LAD that supports the direct instruction of SRL strategies, showing the value of actionable, classroom-ready visualizations for everyday teaching. Complementing these perspectives, Possaghi et al. (2025) integrated multimodal learning analytics dashboards in K-12 contexts, revealing how affective, behavioral, and contextual data can enhance orchestration and teacher decision-making while also raising challenges of workload and interpretability. Together, these contributions expand the evidence base for LAD adoption in K-12 and highlight how dashboards can move beyond monitoring toward actively

shaping pedagogy and classroom practices. Unlike these tools, the MLADs we developed integrate pedagogical frameworks of formative assessment and differentiated instruction directly into their design. By combining multiple data streams (digital learning materials, attendance, and grades) with predictive analytics, our dashboards go beyond descriptive reporting to provide actionable insights for teachers. This alignment with teachers' daily needs reflects the shift from general-purpose monitoring tools toward co-designed, pedagogically embedded analytics solutions (Ifenthaler & Schumacher, 2016).

Methodology

Research design

The DBR methodology employed in this study consists of four main phases (presented in Fig. 1): (1) identifying the requirements, (2) prototyping and ideation, (3) co-designing and developing, and (4) implementing and evaluating the MLADs in educational environments.

The *first phase* involved conducting interviews with teachers to gain insights into their needs and expectations regarding LADs. According to Siemens (2013), understanding the pedagogical context is essential for the successful integration of LAs in educational practice. These interviews aimed to identify factors influencing the adoption of such tools, as well as the specific data visualizations and functionalities that teachers would find most beneficial (Wise, 2014). Participants were prompted to reflect on their existing teaching strategies, educational theories used, current use of educational technologies, and potential barriers to the adoption of data-driven tools (Patton, 2002). This step also served to increase participants' awareness of the capabilities and potential benefits of LAs, thereby laying a foundation for designing dashboards that align with teachers' needs (Groff, 2013).

Based on the insights gathered from the interviews, the *second phase* focused on the creation of paper prototypes (Smith, 2019). Paper prototypes, often referred to as low-fidelity prototypes, are created using simple materials like paper sketches or other basic resources. They allow designers to efficiently and cost-effectively test and refine design concepts by gathering valuable feedback (Smith, 2019). Prototyping is a key component of DBR as it facilitates early engagement with users, understanding of the product/service at hand, and encourages iterative refinement (Holstein et al., 2019). During this phase, different paper prototypes were developed. We conducted three collaborative workshops in three municipalities, where teachers participated in co-designing the paper prototypes. Using

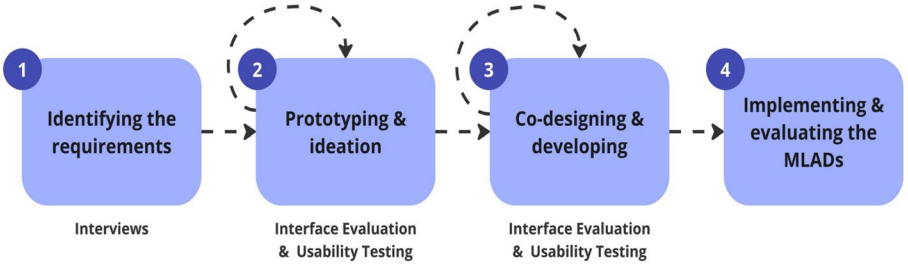


Fig. 1 Proposed four-phase DBR approach

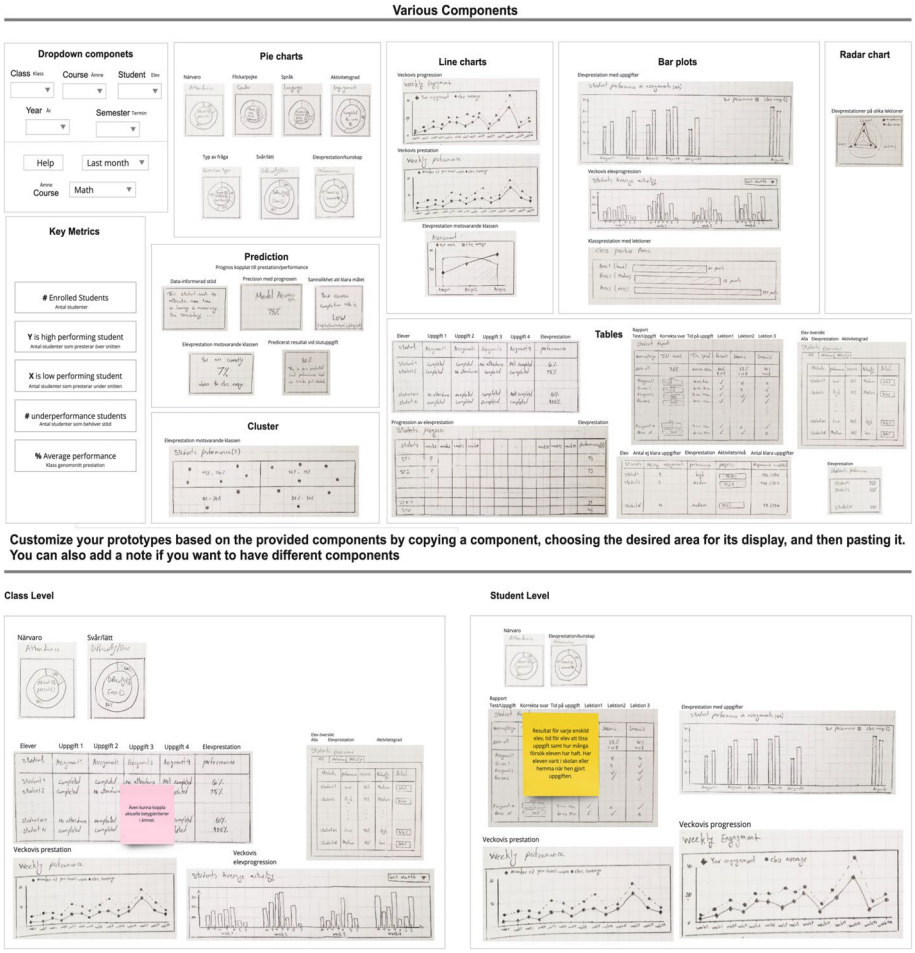


Fig. 2 One example of the Miro board prototyping process for the teacher's LADs

Miro[1] boards for collaborative ideation enabled participants to move and arrange interface components like buttons, drop-down menus, and visualizations in real-time, fostering a shared understanding and collective input (Fig. 2). This design process was instrumental in deepening the understanding of teachers' preferences and identifying visualization features that could enhance the usability of the MLADs (Bodker, 1999).

The *third phase* involved the co-design of interactive prototypes and development of MLADs using insights from the previous phase. Interactive prototypes, often referred to as high-fidelity prototypes, provide a realistic and immersive experience by closely mirroring the final product in both visual design and functionality, allowing users to actively engage with the interface (Smith, 2019). These prototypes underwent 3 iterations, incorporating feedback from usability tests with teachers at each stage. Usability testing, as defined by Rubin and Chisnell (2011), is "a crucial human-centered design method for evaluating the

[1] https://miro.com/

effectiveness and efficiency of user interfaces", aimed at ensuring software is intuitive and effectively fulfills user needs.

The development of the MLADs was guided by comprehensive feedback gathered from teachers during both the interactive prototyping phase and usability testing. This iterative process ensured that the dashboards were refined and tailored to meet the practical needs and preferences of the teachers, aligning closely with their expectations and improving overall usability.

Figure 3 illustrates the iterative workflow of the interactive prototypes across three iterations, along with the enhancements made to one of the developed LADs within our MLADs framework over three iterations. To enhance usability for teachers, the implemented versions of the LADs, referred to as EasyVis dashboards, were translated into Swedish. The figure highlights the progressive improvements achieved in both the interactive prototypes and the developed dashboard. Across iterations (a to c), the LAD prototypes advanced with



Fig. 3 Workflow of interactive prototypes and enhancements to one of the developed LADs within our MLADs framework: **a** prototype—first iteration, **b** prototype—second iteration, **c** prototype—third iteration, **d** dashboard enhancement—first iteration, **e** dashboard enhancement—second iteration, **f** dashboard enhancement—third iteration

improved functionalities, refined visualizations, and enhanced user interfaces to better support insights at both class and student levels. Similarly, in the developed EasyVis dashboard (d to f), iterations focused on optimizing data representation, interactivity, and the depth of analysis, leading to a more comprehensive and user-friendly dashboard. We also conducted a workshop with teachers (described in Sect. "Procedure"), where we presented the latest iteration of the developed EasyVis dashboards.

The teachers completed a 32-question survey in Swedish, with the survey template provided in Appendix B. The results of the survey guided us in improving the three developed EasyVis dashboards for teachers within our MLADs framework and making final adjustments before granting them access to the dashboards.

The *fourth and final phase* involved the implementation and evaluation of the EasyVis dashboards within our MLADs framework in real-world settings across three municipalities. Implementing EasyVis dashboards in authentic educational environments is critical for assessing their practical utility and scalability (Means & Harris, 2013). Teachers tested the dashboards during their regular instructional activities, providing feedback on the usability and effectiveness of the tool in supporting data-driven decision-making (Knight & Shum, 2017). Evaluating the MLADs in situ allowed for a comprehensive assessment of their impact on educational practice, focusing on the utility of the visualizations and the overall user experience (Tsai & Gasevic, 2017).

Teacher feedback from interviews and usability testing sessions was analyzed using formal thematic analysis as described by Braun and Clarke (2006). Initially, all interview sessions and usability tests were audio-recorded and transcribed verbatim. Two researchers then read the transcripts carefully and repeatedly to gain familiarity with the data. Each researcher independently identified and labeled themes and patterns relevant to dashboard usability, functionality, clarity of visualizations, pedagogical alignment, and any barriers teachers encountered. After the independent coding, the researchers compared their identified themes to assess consistency. This consistency, termed intercoder reliability, was measured using Cohen's Kappa coefficient, resulting in a value of 0.82, indicating substantial agreement according to Landis and Koch (1977). A substantial agreement indicates that the researchers largely concurred on interpreting teacher responses, ensuring reliable qualitative analysis results. This thematic analysis directly informed the iterative design process.

For instance, one teacher noted during an early interview: "*It's hard to tell which students are falling behind unless I manually track each result across weeks*". This feedback led to the inclusion of visual trends and predictive alerts to highlight at-risk students in the dashboard. Also, teachers frequently noted that dashboard navigation could be more intuitive, and certain visualizations required simplification or clarification. For instance, a teacher expressed a challenge with the first prototype's layout: *"I get lost in the dashboard…, I don't know where to find the data I need"*. This feedback led to a redesign of the navigation interface for easier access, improved labeling of tabs (e.g., "Class level" and "Student level") and dropdown menus, clearer visual transitions between views, and the addition of interactive elements such as explanatory pop-ups. Similarly, another teacher emphasized the need to unify multiple data sources: "*It would help to have student attendance and DLM performance in the same place*". As a result, we integrated both attendance and performance data into the class-level tab, allowing teachers to assess patterns more holistically.

Additionally, during a later co-design session, one teacher pointed out: "*It's important for me to know where the data comes from. I need to trust what I'm seeing*". This insight prompted the inclusion of data source labels in each dashboard section and the

development of the Help tab, which explains data origins and visualizations in detail (see Figs. 6 and 7). These insights, gathered through co-design sessions and interviews throughout the DBR process, were essential in informing the visual and functional aspects of the EasyVis dashboards, with explicit design modifications clearly documented to show how each iteration directly responded to teacher feedback.

To make the iterative structure of the DBR process more explicit, Table 1 summarizes the four main phases according to Sandoval's (2014) conjecture–embodiment–warrant model. Each phase shows how design ideas were successively implemented, evaluated, and refined based on teacher feedback and empirical findings.

Participants and intervention

We conducted an intervention involving 25 teachers across three municipalities in Sweden over a one-year period, from early October 2023 to mid-November 2024 (see Fig. 4). Table 2 provides details regarding the participants involved and the school subjects included. The teachers were distributed as follows: Municipality 1 with 7 teachers, Municipality 2 with 12 teachers, and Municipality 3 with 6 teachers. The participants were not selected and represented all teachers of Mathematics, Swedish, and Swedish as a second language of grade 4–6 in the participating schools. Notably, our collaboration with these teachers began earlier, in January 2022 (Masiello et al., 2023), where we initially focused on their professional development for effective DLM usage. In addition to the 25 teachers, two pedagogical experts, three data and computer scientists, municipal IT administrators, and representatives from EdTech companies were involved.

Procedure

The timeline shown in Fig. 4 illustrates the development process spanning from October 2023 to November 2024 and involves iterative phases of data analysis, design, prototyping, usability testing, feedback collection, and implementation. In October 2023, the project begins with the identification of interview questions and survey. All the questions for the interviews and the 32-question survey—including the 9 interview questions from the first phase—were developed collaboratively through meetings involving three data and computer scientists, and two pedagogical experts. We developed our own questions to ensure that they reflected the pedagogical frameworks central to this study, such as formative assessment and differentiated instruction, as well as the practical needs of teachers in primary education. The questions were co-designed with pedagogical experts, piloted during the first round of interviews, and refined based on teacher feedback to maximize clarity and validity. The questions in the first interview focused on teaching and learning processes, aiming to understand how teachers interacted with students and sought to gain deeper insights into their development. This information was intended to gather valuable perspectives on teaching practices that could guide the development of potential visualizations. This is followed by the analysis of student data collected from EdTech companies and municipalities, which play a crucial role in informing the initial stages of design. During November 2023, a more detailed examination of the collected data is conducted to better understand the context and existing challenges in the educational environment. This phase also includes the first series of interviews with teachers aimed at connecting pedagogical strategies to the design of the LADs. A set of questions used in this series of interviews is outlined in Appendix A.

Table 1 Summary of DBR phases according to the conjecture–embodiment–warrant model

DBR Phase	Conjecture (Design Idea)	Embodiment (Implementation)	Warrant (Evidence / Outcome)
1. Identifying the Requirements	Understanding teachers' needs and pedagogical contexts is essential for designing effective dashboards	Conducted interviews with teachers in three municipalities to identify needs, expectations, and barriers regarding LADs; explored current use of educational technologies and potential visualization needs	Teachers emphasized fragmented data access and requested a unified overview combining attendance, DLM engagement, and performance data to support classroom decision-making
2. Prototyping and Ideation	Early, low-fidelity prototypes can help visualize teachers' ideas and facilitate co-design	Developed paper prototypes (low-fidelity prototypes) and conducted three collaborative workshops using Miro boards for arranging interface components and visual elements	Teachers identified preferred visualization types, simplified navigation, and emphasized usability and clarity of indicators; findings informed the next iteration of interactive prototypes
3. Co-Designing and Developing	Iterative co-design and usability testing ensure dashboards are pedagogically aligned and user-friendly	Created interactive prototypes (high-fidelity prototypes) and developed MLADs through three iterations, integrating real municipal data, adding explanatory pop-ups, and refining three tabs: Class level, Student level, and Help	Usability tests and workshops showed improved clarity, alignment with teachers' practices, and trust in data sources; requests led to inclusion of data source labels and transparency features
4. Implementing and Evaluating	Implementation in authentic contexts validates design and identifies scalability challenges	Deployed EasyVis dashboards across three municipalities; teachers used them during regular instruction and completed a 32-question survey and interviews for evaluation	Teachers valued integrated data views and enhanced awareness of student progress; scalability challenges emerged due to variations in local DLM systems and data formats

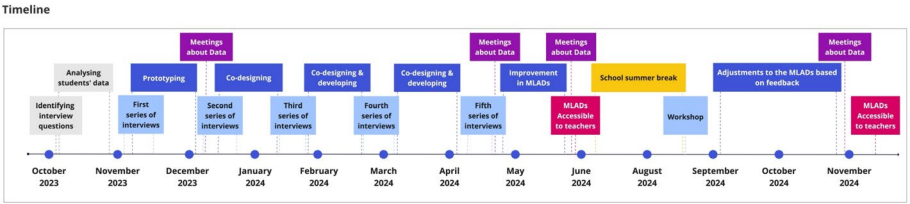


Fig.4 Project timeline indicating the teachers' engagement in the process of co-designing and developing the MLADs

Table 2 Municipality and school participants (table reproduced from Nordmark et al., 2024, used with permission)

Participants		Municipality 1	Municipality 2	Municipality 3
School Leaders involved (Municipality Level)		1	1	1
Tech & Admin Staff (Municipality Level)		1	1	1
CEO Involvement from EdTech Companies		1	2**	2**
Tech Staff Involvement from EdTech Companies		1	2**	2**
No. of Schools		2	1	1
Teachers	Mathematics	4	3	2
	Swedish	3	3	3
	Swedish as a 2nd Language (SweA)	1*	2	1
	Special Education	–	4	–
	Total	*7*	*12*	*6*
Pupils	Grade 4	16	21	48
	Grade 5	26	32	35
	Grade 6	38	28	37
	Total	*80*	*81*	*120*
No. of Participating Pupil (Swedish)		10	31	13
No. of Participating Pupils (with other Backgrounds than Swedish)		21	74	17

**Same person teaching both Swedish & SweA*

**Two municipalities implemented two different DLMs, with one of them being shared between both municipalities

Because it was crucial to raise awareness among school principals, we requested their attendance at the interviews to ensure they understood how the DLMs were being utilized in their schools, to gain a better understanding of teachers' needs or difficulties, and to learn more about how students were using the tools in the classroom. Each series of interviews consisted of three meetings, with each meeting focusing on a group of teachers from one subject (Mathematics, Swedish, or Swedish as a Second Language). This subject-specific structure allowed teachers to express pedagogical needs related to their discipline—for instance, Mathematics teachers prioritized progression tracking, while language teachers emphasized vocabulary and comprehension. These insights were used to identify suitable

visualizations and ensure that the LADs could be adapted to different subject requirements. Following these interviews, the project team collaborates with teachers through co-design sessions using tools like Zoom and Miro, leading to the development of paper prototypes (Fig. 2). By December 2023, the co-design efforts are expanded, and the initial paper prototypes are refined based on feedback from the second series of interviews. This iterative process is essential for integrating the insights gathered from teachers into the design.

Throughout the co-design process, teachers were not merely consulted but actively engaged as co-creators. Each workshop involved groups of local teachers from Mathematics, Swedish, and Swedish as a Second Language, working alongside the research team. Using Zoom and interactive Miro boards, teachers collaboratively arranged interface elements such as charts, filters, and student lists, and gave input on terminology, data presentation, and usability. These sessions enabled them to visually and verbally express both pedagogical needs and design preferences. The co-design team also included two pedagogical experts and three data and computer scientists. The pedagogical experts ensured that proposed visualizations aligned with curriculum goals and instructional strategies, while the data and computer scientists iteratively refined the designs based on teacher feedback. This structure ensured that each co-design session was both practically grounded and technically actionable. As mentioned before, all interview sessions (including co-design workshops) were audio-recorded and transcribed verbatim, complemented by structured observation notes taken in Miro boards. After each round, the research team met to analyze the collected data, categorize teacher feedback into themes, and translate these into concrete design actions. This process directly led to modifications such as adding predictive alerts, simplifying navigation structures, integrating attendance and performance data in one view, and developing a Help tab to explain data sources and visualizations. These design changes were then implemented and validated in the subsequent prototype iterations. Feedback was further validated through multiple prototype test cycles and usability testing, which informed each major refinement in the dashboard iterations (see Fig. 3).

The focus then shifts to co-designing interactive prototypes from mid-December to mid-January 2024. Also, throughout this phase, we conducted the first series of meetings with EdTech companies and municipalities to ensure the accuracy of their data and discuss the integration of DLMs. In January 2024, the interactive prototypes are tested in the field. The third round of interviews with teachers is conducted to evaluate the prototype's effectiveness and gather insights on its practical application in classrooms. This feedback informs further improvements to the interactive prototypes, ensuring that it aligns with teachers' needs. In February 2024, leveraging the insights gained from these evaluations, the project team develops the first version of the EasyVis dashboards within our MLADs. A fourth round of interviews with teachers and school principals is conducted to gather comprehensive feedback on the initial version. This phase focuses on identifying areas for improvement and planning for the second iteration. Alongside these interviews, the first round of usability testing is conducted with 8 Mathematics teachers, 7 Swedish teachers, and 3 Swedish as a Second Language teachers via Zoom. Teachers are asked to control the dashboards remotely and perform typical tasks such as identifying struggling students, test different functions, and share their thoughts while researchers and pedagogical experts observed and documented difficulties, preferences, and usability issues. Usability was evaluated through observation notes, and follow-up discussions. By March, development begins on the second version of MLADs, which includes enhancements based on the feedback received. The project progresses to March 2024 with final evaluations of the second version of the MLADs. This phase marks a crucial milestone where improvements are integrated into the dashboards, making them more user-friendly for teachers. By April 2024,

the refined second version is further tested and adjusted. During this time, meetings with EdTech companies and municipalities continue to refine data integration and ensure the dashboards are adapted to real-world educational contexts. In April 2024, a fifth round of interviews is conducted to assess the improved MLADs. At this stage, the second round of usability testing is carried out on-site in the three schools, involving 8 Mathematics teachers, 8 Swedish teachers, and 4 Swedish as a Second Language teachers. Teachers explored the updated dashboards under observation, complete representative tasks, and discuss their experiences. Feedback is collected through observation notes, and follow-up interviews. Based on this combined input, additional adjustments are made in May 2024, ensuring that the third version of the dashboards are ready for broader accessibility to teachers by June 2024. Teachers are invited to explore and interact with the dashboard and share feedback during the meeting. The project enters a pause during the summer break from June 12 to August 17, 2024, when schools are closed. However, planning and strategizing for the next phase continue. In August 2024, workshops are held with teachers to assess the practical implementation of the dashboards in preparation for the upcoming academic year. This third round of usability testing includes 9 Mathematics teachers, 9 Swedish teachers, and 4 Swedish as a Second Language teachers. The sessions are organized as subject-specific workshops, where teachers work in groups to explore the dashboards, and provide feedback. At the end of the workshop, the teachers completed a 32-question survey in Swedish (Appendix B). Usability is evaluated through the survey, observation notes from feedback sessions, and interviews. By September–October 2024, further adjustments are made to the MLADs based on feedback from these workshops and additional modifications are integrated into the dashboards, informed by surveys and interviews conducted with teachers. Continued collaboration with EdTech companies and municipalities ensures that the data remains relevant and useful for educational settings. By November 2024, the final, enhanced version of the MLADs is launched, now fully accessible to teachers through a secure login system. This release represents the culmination of over a year of iterative development, collaborative input, and continuous refinement. The result is a robust set of dashboards specifically designed to address teachers' needs and enrich the learning process with actionable, data-driven insights. In line with this objective, we did not collect usage logs from the developed MLADs, as its' purpose is to support teachers' pedagogical decision-making rather than to monitor their behavior. Nevertheless, the MLADs framework has the technical capacity to record usage data, and this remains an avenue for future research. The present study therefore relies primarily on self-reported and observational data, which we acknowledge as a limitation.

Ethical considerations

All procedures in this study complied with the European Union's General Data Protection Regulation (GDPR) (Voigt & Von Dem Bussche, 2017) and the Swedish Government Act 460 (2003) on the Ethical Review of Research Involving Humans. Ethical approval was granted by the Swedish Ethical Review Authority, in line with the framework applied in related national research on educational technology implementation in Swedish K-12 contexts (Nordmark et al., 2024). The study was carried out in collaboration with three Swedish municipalities that provided anonymized datasets for dashboard development. The data consisted only of educational variables (such as attendance records, DLM engagement, and student assessment outcomes) necessary for testing the dashboard functions. No sensitive personal information such as ethnicity, health status, or socioeconomic indicators

was included. All participating teachers provided informed consent before engaging in co-design workshops and dashboard testing. Written guardian consent was obtained for the use of anonymized student data in accordance with municipal data-management policies. The predictive models embedded in the MLADs were trained only on de-identified data and were not connected to live systems. No usage logs from the MLADs framework were collected. This decision was made to avoid monitoring teachers' actions or classroom behavior and to uphold the voluntary nature of participation. Although this choice limited opportunities for detailed interaction analysis, it supported the ethical principles of transparency, professional autonomy, and mutual trust between researchers and practitioners. These measures ensured full compliance with GDPR and Swedish ethical requirements, emphasizing fairness, accountability, and responsible handling of educational data throughout the DBR process.

Outcomes

As part of the iterative design process, three EasyVis dashboards within our MLADs framework were developed, each customized to meet the specific needs of teachers in each municipality. Each dashboard integrated various visual components such as bar charts, pie charts, line graphs, text boxes, and tables aimed at presenting actionable data. Figures 5, 6and 7 illustrated the latest iteration of the LAD developed for the teachers in one of the municipalities as an example. It is important to highlight that there are variations between the MLADs, again due to the available data by the EdTech providers and municipalities.

In this version of the EasyVis dashboard, we combined data from students' learning activities collected by an EdTech company, with attendance data and final grades provided by a municipality. This integration allowed for a comprehensive assessment of student engagement, performance, attendance, and prediction. The dashboards were designed to clearly display the source of the data within each section to inform teachers about the origin of each visualization's data, aiming to build a sense of trust between the user and the dashboard. All three dashboards in our MLADs included a secure login system and three tabs: *Class level*, *Student level* and *Help*.

Figure 5 shows the welcome and login pages. All teachers were assigned a link to the school dashboard and a complex password that was automatically generated and could not be changed into a simpler one. If a teacher entered an incorrect username or password, a warning page appeared, prompting them to try again. Figure 6 presented data at both the *Class* and *Student* levels.

Class level

The *Class level* tab at the top of Fig. 6 provides insights into the composition of students within a classroom.

Teachers can select a grade, a semester and a course name to access general classroom statistics, such as the number of students and the distribution of high- and low-performing students. Pie charts display the breakdown of students by gender, their performance, shown as the most recent grade, and their last registered assessment. Additionally, a bar chart in the upper part of the Class level tab shows students' weekly performance in the DLM. This is determined by the correct and incorrect answers to various assignments in the DLMs, or the frequency of interactions with the DLMs. A table provides more detailed information

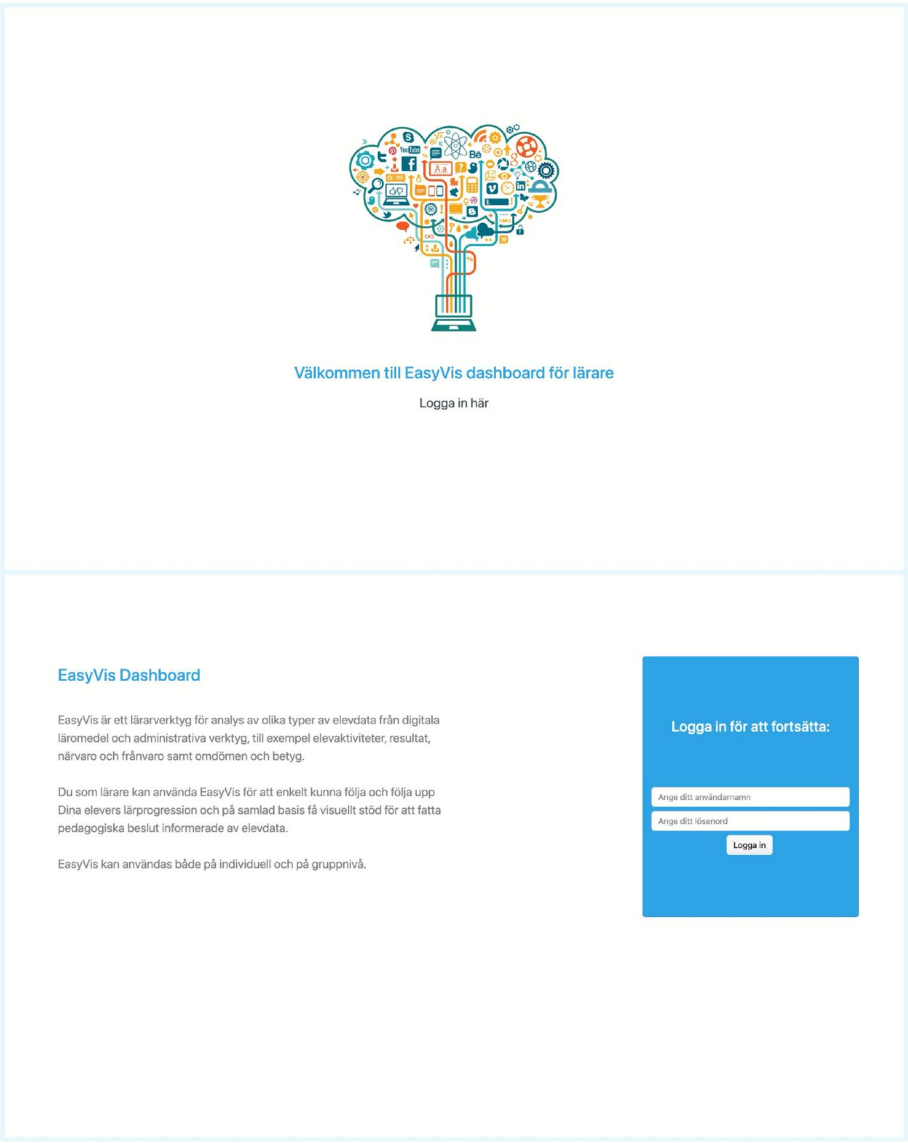


Fig. 5 The developed EasyVis dashboard for teachers: Welcome and login pages

on student IDs, student attendance throughout the semester, gender, attendance percentages, and their most recent assessment. A bar chart in the lower section of the tab allows for easy comparison of individual student scores with the class average, helping teachers assess whether a student's performance is above or below the class average. Moreover, teachers can adjust a desired benchmark value as the "average", according to the trend performance of the whole class. By clicking on a bar in the chart, teachers can view predictions about the selected student in the top right of the tab. To generate these predictions, we incorporated machine learning into the dashboards. The algorithms analyze historical

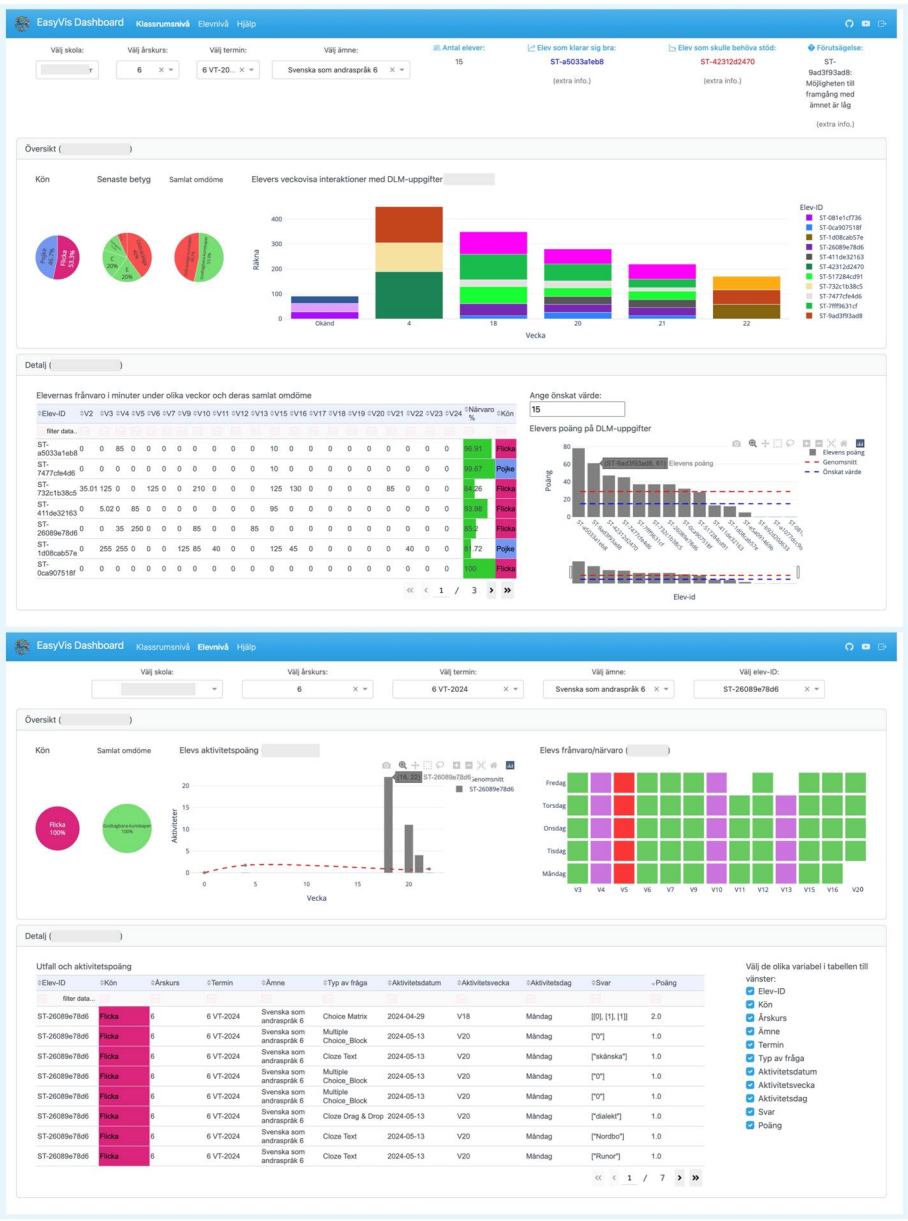


Fig. 6 The developed EasyVis dashboard for teachers in the first municipality: Class and Student tabs

data to forecast student performance, attendance and scores, providing proactive insights such as early alerts for students at risk of falling behind. To make the predictive process transparent, our MLADs framework incorporated a Random Forest Classifier (Breiman, 2001) that generated early alerts regarding course completion risk. The model analyzed aggregated patterns of attendance, DLM activity, and assessment performance over weeks.



Fig. 7 The developed EasyVis dashboard for teachers in the first municipality: Help tab

A custom classification function categorized each student's likelihood of successful course completion into three levels: low risk (1), medium risk (2), and high risk (0). This classification was based on comparisons between each student's accumulated points and the class average (students performing within one point of the class average were labeled as having medium completion risk, those above the average as low risk, and those below as high risk). The model was trained on 70% of the available historical data and validated on the remaining 30% using five-fold cross-validation. The evaluation yielded an F1-score of 0.82 and an area under the ROC curve (AUC) of 0.89, which were considered satisfactory for the intended formative use (Jeni, Chon & De La Torre, 2013). These predictions were displayed as advisory indicators within the Class level tab, providing teachers with an overview of students' relative progress. Importantly, the predictive function was designed to support (not replace) teacher judgment, consistent with recommendations for responsible and interpretable learning analytics (Datnow & Hubbard, 2016; Ifenthaler & Schumacher, 2016).

By automatically consolidating grades, attendance, and DLM engagement into a single view, and by offering predictive alerts, the dashboards reduced the amount of time teachers previously spent manually tracking and cross-checking information, addressing the barrier of limited time identified in the introduction.

Student level and help

In the *Student level* tab, shown in the lower part of Fig. 6, teachers can select a student ID from a dropdown menu to access more detailed information about that student, such as their most recent assessment, gender, and engagement across different weeks of the school semester in comparison to the class average. The student's attendance status is also displayed with green indicating presence, purple indicating partially present, and red indicating absent. A table shows the dates of the student's engagement in different lessons within the selected grade, course and school semester. As students do not use the DLM weekly, the visualizations only display weeks when they were actively engaged with the platform.

Finally, the *Help* tab, shown in Fig. 7, provides additional information about the LAD, the data, its origin, and the various visualizations used in the dashboard. Again, the description of the origin of the data is necessary to build trustworthy LADs. This functionality also supported teachers with different levels of data literacy by providing clear explanations of each data source and visualization, thereby reducing the barrier of data interpretation highlighted in the introduction.

Feedback from teachers

This article does not focus on showcasing the qualitative data from these interviews (though a few insights are included in Sect. "Research design"); rather, it emphasizes the DBR methods employed and highlights the findings derived from them. In August 2024, we conducted a workshop attended by all 25 teachers, three school principals, three municipality education leaders, three IT-administrators at the municipalities, and four EdTech company representatives, where we demonstrated and let all participants test for the first time the MLADs. At the conclusion of the workshop, 18 of the 25 teachers completed a 32-question survey in Swedish. This feedback provided valuable insights into how effectively the MLADs could perform in real-world educational environments. Figure 8 represents five pie charts and one bar plot indicating user responses to Q1-Q6 in the survey



Fig. 8 User responses to Q1-Q6 in the survey: **a** gender, **b** age category, **c** level of education, **d** teaching experience, **e** teaching subject, **f** knowledge about DLMs

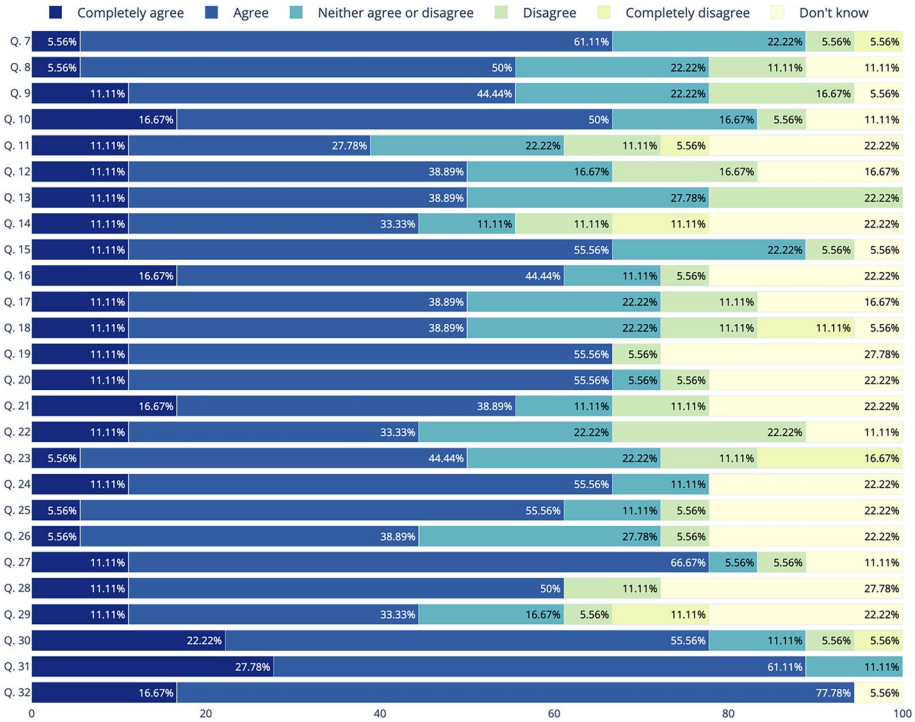


Fig. 9 User responses to Q7-Q32 in the survey

(Appendix B) related to demographic information, and user experience. Figure 9 shows the distribution of answers to questions Q7-Q32 in the survey. These questions covered different aspects of the LADs' usability, such as clarity, navigation, functionality, and overall user experience.

Figure 8a focuses on gender distribution, showing that the majority of respondents are women (77.8%), while men account for 22.2%. Figure 8b illustrates the age categories, with the largest proportion (44.4%) falling into the 41–45 age group. Equal representation (11.1%) is observed among age groups 31–35, 46–50, and 61–65, while younger age groups, such as 20–25 and 26–30, have minimal representation. Educational attainment is depicted in Fig. 8c, where 72.2% of participants hold a university degree, 11.1% each have either a bachelor's degree or vocational education, and 5.6% have some college credits, reflecting highly educated participants. Figure 8d examines teaching experience, revealing that 38.9% have over 20 years of experience. A significant portion (27.8%) has less than 5 years, while 16.7% have either 5–10 years or 16–20 years of experience. The subjects taught by teachers, shown in Fig. 8e, highlight mathematics and Swedish as the most commonly taught subjects, with 10 and 9 teachers, respectively, identifying them as their specialty. A list of other subjects follows.

Lastly, Fig. 8f addresses teachers' familiarity with DLMs. Most participants (72.2%) describe themselves as familiar, while 27.8% identify as very familiar, suggesting a generally high level of comfort with digital interfaces, though fewer exhibit advanced expertise. This is also thanks to the results of the larger implementation project for which this study is part of (Masiello et al., 2023).

The stacked bar chart in Fig. 9 illustrates the distribution of responses to 27 survey questions (Q7-Q32) about the usability, clarity, design, and accessibility of the MLADs. Each question was rated on a Likert scale ranging from "Completely agree" to "Completely disagree" and "Don't know". Here's a detailed analysis of all responses.

Strengths

- *User Familiarity with Digital Tools*: The survey results indicate that teachers are largely comfortable with the DLMs. Specifically, Q7 ("Knowledge about digital learning tools") shows a high level of confidence, with the majority expressing strong agreement.
- *Text Clarity and Readability*: The LADs perform well in areas related to text presentation, such as the wording of content (Q26), ease of understanding (Q27), and font readability (Q31, Q32).
- *Information Display and Labeling*: Respondents agree that labels and headings are clear (Q16) and that most visual components perform as expected (Q11, Q13). This indicates that teachers generally find the layout and information structure to be intuitive and easy to follow.

Areas for improvement

- *Navigation and Consistency*: Despite the positive feedback on general clarity, there are concerns around navigation (Q9, Q15) and consistency of interactive elements (Q17, Q18). Improving consistency and ensuring that navigation is straightforward should enhance overall usability.
- *Warnings for Critical Actions*: A significant proportion of teachers expressed dissatisfaction with the lack of clear warnings before performing irreversible actions (Q20).

- *Accessibility of Supplementary Information*: Questions Q28 and Q29 highlight challenges in accessing supplementary or "extra info" content. Teachers either found this information difficult to locate or were unaware of its existence. Content should be made more visible and easier to access.
- *Clarity in Inputs and Form Fields*: Teachers reported some difficulty in understanding what information was required in certain input fields (Q22). This could indicate that the LADs' form components and dropdown menus may not be adequately labeled or intuitive.

Neutral or uncertain feedback

Several responses (Q10, Q12, Q23) showed a higher proportion of neutral or "don't know" answers. This could indicate that certain aspects of the LADs' design, such as the logical flow of information or the balance of content, are not immediately apparent to teachers.

Taking into account the comprehensive feedback gathered from the survey in August 2024, and our discussions with each municipality and associated EdTech companies afterward, we implemented a series of adjustments during September and October 2024, focused on enhancing various tabs, ensuring improved functionality, usability, and alignment with teachers' needs. The modifications reflect our commitment to incorporating stakeholder input and delivering a more effective and user-friendly platform.

Challenges in scalability

As the study progressed, it became clear that the scalability of the MLADs across different municipalities was a significant challenge. Each municipality had different educational data sources, and the learning management systems (LMS) used by each municipality varied. This made it difficult to create a one-size-fits-all solution for all municipalities. The LADs within our MLADs framework were designed to be as scalable as possible, but the varying data sources required customization in the dashboards to ensure that the data presented was relevant to each specific educational context. This challenge was compounded by the fact that teachers in different municipalities often had different levels of familiarity with data-driven dashboards. Teachers in some municipalities were more comfortable using technology to track student performance, while others were less experienced. This variation required the research team to offer targeted support to ensure that all teachers could effectively use the MLADs (Nordmark et al., 2024). This project remains ongoing, although preliminary findings have been obtained.

To illustrate how modularity supported implementation across municipalities, Table 3 summarizes the MLADs components that remained consistent and those that required local adjustment. The modular structure meant that core design elements (such as the dashboard architecture, tab layout, and visualization types) were reused without modification, while data integration and terminology adjustments were tailored to local systems. Although the technical and visual frameworks were identical, each municipality required separate validation of the interface labeling because the DLMs differed across municipalities and provided data in different formats. The adaptation process mainly involved verifying data compatibility, adjusting field mappings, and testing the dashboards to ensure that indicators were correctly represented. This work required a few days per municipality, primarily to recheck the code and data flow rather than to redevelop components. Overall, these findings suggest that the modular design of the MLADs enabled relatively efficient scalability while still

Table 3 Shared and adapted MLADs components across municipalities

Dashboard Component	Common Across Municipalities	Locally Adapted Elements
Core architecture and framework	Shared MLADs base using Python/Plotly Dash	None
Tabs and structure	Login system, Class level, Student level, and Help tabs, dropdown components	None
Visualization types	Bar charts, pie charts, tables, heatmaps, predictive-alert text, pop-ups	Minor adjustments reflecting teaching subjects
Data sources	Student information system integration	DLM connections differed
Assessment data	Grades, attendance, and engagement indicators	Different grading scales (A–F vs points)
Terminology and labels	Standard Swedish pedagogical terms	Minor adjustments reflecting local curriculum wording
Color scheme / interface theme	Shared default layout	Adjusted to municipal branding in some cases
Predictive component	Random Forest-based alert model	Same model structure, retrained with each municipality's dataset

requiring careful contextual alignment. The replication effort focused more on ensuring data integrity and consistency than on rewriting code, showing that modular structures can support transfer across contexts when local data infrastructures and content providers vary.

Discussion

This study aimed to co-design, develop and test MLADs to support data-informed decision-making among teachers in primary education. Through an iterative design process, the research addressed critical issues related to the co-design, usability, functionality, and scalability of these dashboards. The results demonstrate that co-design is a very important step for developing MLADs that can provide valuable insights into student performance, engagement, and progress. Although, several challenges emerged, embedded below.

Co-design of MLADs

Research has shown that collaborative design methods can significantly enhance the relevance and usability of educational tools by involving end-users in the design process (Sanders & Stappers, 2008; Smeenk, 2023). The co-design process in this study facilitated active collaboration between researchers and teachers, ensuring the MLADs were both contextually relevant and user-centered. Such approaches are supported by prior work demonstrating the effectiveness of iterative design and refinement in educational technologies (McKenney & Reeves, 2018). Co-design process was crucial for refining the MLADs to ensure that they were easy to navigate and effective in terms of delivering meaningful visualizations and interactions (Sauro & Lewis, 2016). By prioritizing user experience design (UXD), this step aimed to create a dashboard that was both functional and accessible, addressing the diverse requirements of teachers (Norman, 2013). The findings highlight that involving teachers in co-design not only improved usability but also empowered them to take ownership of the technology, echoing similar successes reported in Holstein et al. (2019). This collaboration allowed for the identification and prioritization of features such as simplified navigation, tailored data visualizations, and alignment with pedagogical practices. The iterative prototyping approach, from low-fidelity paper designs to high-fidelity interactive dashboards, mirrors best practices in human-centered design (Norman, 2013; Rubin & Chisnell, 2011).

However, co-designing MLADs posed several challenges. Balancing the diverse perspectives of stakeholders required careful negotiation, as teachers often prioritized usability while researchers focused on theoretical alignment (McKenney & Reeves, 2018; Sanders & Stappers, 2008). Additionally, differences in institutional contexts, resource availability, and technological familiarity complicated the creation of universally applicable solutions, as has been noted in the study by Robertson and Simonsen (2012). Finally, effective communication was another challenge, as limited time and technical expertise among some participants occasionally hindered engagement (Nicholson et al., 2022).

Usability and functionality of MLADs

The positive reception of the MLADs by teachers regarding their usability and design suggests that the dashboards have great potential to be used by teachers and possibly in making informed decisions about student progress. The findings from this study align with

prior evidence (Masiello et al., 2024), which emphasizes that the design of educational tools should prioritize ease of use and intuitive navigation. In particular, the clear starting page and defined subpages were highlighted as critical features that contributed to the tool's usability, mirroring findings in previous studies that stress the importance of intuitive interfaces in educational technology (Bodily et al., 2019; Ifenthaler & Schumacher, 2016).

Nevertheless, while the usability was generally well-received, some teachers found that the data organization within the dashboards could be improved. This finding suggests that there remains room for refinement in the data presentation to ensure that it is logically organized, particularly when considering different user types and educational contexts. The challenge of organizing complex data in a user-friendly way has been discussed in previous literature (Liu et al., 2017), where researchers have highlighted the difficulty of presenting large datasets in ways that are accessible to non-expert teachers. This indicates a need for further research into how data can be structured to better support user understanding, particularly for teachers with varying levels of experience with data analytics. This points out also to the need for data-literacy (Kippers et al., 2018).

Scalability across different educational contexts

A significant challenge identified in this study was the scalability of the MLADs across different municipalities with varying educational data sources and teaching practices. As mentioned, the municipalities involved in the study used different DLMs and had varying approaches to data collection and structure, which made it difficult to develop a one-size-fits-all dashboard. This observation highlights a key limitation of current educational technology solutions, as they often face challenges in adapting to diverse educational contexts (Batini et al., 2015). The variability in data sources requires that dashboards be highly customizable to meet the specific needs of different schools or regions, a concern echoed by other studies on the challenges of scaling educational technologies (Govea et al., 2023). While this customization ensures the relevance of the data, it also increases the complexity of the design and implementation process, which can be a barrier to widespread adoption.

Furthermore, the variation in teachers' familiarity with technology across different municipalities also posed a challenge. In some municipalities, teachers were already familiar with using technology to track student performance, while in others, there was less experience with DLMs for educational management. This disparity reflects findings from previous research, which highlight the importance of considering user experience when implementing technology in education (Liu et al., 2017). Therefore, while the MLADs were well-received in municipalities where teachers had experience with data tools, further efforts are needed to provide support and training in schools where teachers are less familiar with using technology in educational decision-making.

This study contributes to the DBR literature by demonstrating how modular design elements within learning technologies can support scalability while preserving local relevance. The MLADs evolved not as one-size-fits-all tools, but as configurable frameworks grounded in teachers' pedagogical goals and institutional constraints. We propose a modular dashboard architecture as a practical instantiation of DBR principles—where core components (e.g., student-level progress views, class summaries) are reusable, while peripheral elements (e.g., data source connectors, assessment labels) are tailored to the specific educational context. For example, in one municipality the DLM data included students' scores, which we visualized using a pie chart, whereas in another municipality the same component was adapted to display the number of correct and incorrect answers. Similarly, the

table visualization was reused across municipalities but adjusted to reflect the specific data available in each local system. These examples illustrate how modularity allowed MLADs to be reused across contexts while still being flexibly adapted to local conditions. In this way, our findings extend McKenney and Reeves' (2018) notion of DBR scalability, which emphasizes iterative refinement and contextual adaptation, by showing how modularity can serve as a concrete design mechanism that enables such adaptation across diverse educational infrastructures. This modularity facilitates scaling without compromising fidelity, offering a valuable theoretical contribution to DBR models that seek to bridge localized design with broader educational impact (McKenney & Reeves, 2018). At the same time, we caution that the evidence presented here is limited to three municipalities, and further studies across different educational systems are required before claims of systemic impact can be substantiated. In future iterations, we envision a component-based authoring system that allows new schools or municipalities to configure their dashboards by selecting from a library of validated modules. Such a system aligns with recent calls in the literature to design for adaptability rather than static replication (Campanella & Penuel, 2021).

Beyond scalability, the sustainability of MLADs is critical for long-term impact. In our project, municipalities and EdTech partners committed to maintaining the dashboards beyond the research phase, including responsibility for updates to ensure technical compatibility with evolving DLMs and school information systems. Sustainability also depends on continuous teacher training, as turnover and new staff require support to effectively use the dashboards. Finally, financial sustainability requires municipalities to allocate resources for ongoing maintenance and support. These measures indicate that while the MLADs are scalable, their sustained use depends on strong institutional ownership, regular training, and stable funding structures.

Barriers revisited: progress and remaining challenges

Returning to the barriers identified in the introduction, the findings suggest mixed progress. The barrier of limited teacher time was partly addressed since predictive alerts and the integration of grades, attendance, and DLM activity helped teachers act more efficiently, although time constraints remain a structural challenge. As one teacher explained, "*It saves me time because I don't have to check attendance and assignments separately anymore. It's all in one place*". The barrier of low data literacy was partly alleviated because simplified visualizations, labeling, and the Help tab supported teachers in interpreting the dashboards, but further training is still required to ensure consistent adoption. One teacher emphasized this point, noting, "*Seeing exactly where the data comes from makes me more confident in using the dashboard with my class*". The barrier of trust and usability was largely overcome as co-design, transparent data labeling, and iterative testing enhanced teachers' confidence in the dashboards, even though some navigation challenges remain. Taken together, these findings show that MLADs can reduce certain barriers, but their long-term success depends on sustained professional development and institutional support to embed data-informed practices into everyday teaching.

Teacher professional development and support

The use of MLADs also contributed to teachers' professional development, as they were exposed to new ways of understanding and interpreting student data. However, while teachers appreciated the data-informed insights provided by the dashboards, some struggled

with understanding how to incorporate these insights into their day-to-day teaching practices. For instance, studies by Datnow and Hubbard (2016) highlight that teachers face difficulties in translating data into actionable pedagogical strategies, largely due to challenges in data literacy and interpretation. This points to the need for targeted professional development that focuses on the pedagogical use of LAs in the classroom.

Transferable design principles for MLADs development

Drawing on the iterative DBR process, co-design activities, and implementation of the MLADs across three Swedish municipalities, several transferable design principles emerged that can inform future LAD development and research in primary education and related contexts. These principles synthesize what was learned across cycles of conjecture, embodiment, and warrant (McKenney & Reeves, 2018; Sandoval, 2014).

1. *Ensure transparency and data provenance to strengthen trust*: Each LAD clearly displayed the origin of data sources and provided explanatory help sections describing how information was processed and visualized. This transparency supported teachers' confidence in the dashboards and reduced uncertainty regarding the validity of the displayed data, consistent with findings from previous research emphasizing trust and interpretability in learning analytics (Ifenthaler & Schumacher, 2016; Viberg & Grönlund, 2021).
2. *Adopt a modular and adaptable architecture to enable scalability*: The MLADs shared a common structure including three tabs (Class level, Student level, and Help), while allowing localized adaptations in data integration and visualization to accommodate differences in municipal systems. This modularity demonstrated that scalable solutions can retain local relevance through minimal configuration changes (McKenney & Reeves, 2018).
3. *Align visualizations with pedagogical frameworks*: The MLADs operationalized principles of formative assessment and differentiated instruction by visualizing performance, engagement, and attendance data in ways that directly supported teachers' instructional decision-making (Black & Wiliam, 2009; Tomlinson, 2014). Linking analytics to pedagogical practice ensured that the dashboards were actionable rather than merely descriptive (Ifenthaler & Yau, 2020).
4. *Simplify visual complexity to support data literacy*: Across iterative prototypes, visualizations were progressively simplified, redundant indicators were removed, and language was adjusted to plain Swedish. These refinements reflected teacher feedback and ensured that users with varying levels of data literacy could interpret the dashboards efficiently (Kippers et al., 2018).
5. *Use predictive analytics as supportive cues rather than prescriptive judgments*: Predictive alerts were implemented to identify students potentially at risk of underperformance but always presented as advisory indicators for reflection rather than automated decisions. This design respected teacher autonomy and reinforced the view of analytics as a complement to professional judgment (Datnow & Hubbard, 2016).
6. *Embed continuous co-design and reflection throughout DBR cycles*: Sustained collaboration with teachers, pedagogical experts, IT staff, and EdTech partners across iterations ensured that design choices were grounded in classroom realities and pedagogical goals (Holstein et al., 2019; Sanders & Stappers, 2008). Iterative feedback loops enhanced

usability, fostered teacher ownership, and ensured that theoretical conjectures were continually refined through practice.

Together, these principles illustrate how DBR can bridge the gap between theory and practice in LA design. They offer a concise synthesis of actionable guidance for researchers and practitioners developing MLADs or similar educational technologies, emphasizing transparency, adaptability, pedagogical alignment, and sustained collaboration as foundations for scalable, trustworthy, and pedagogically meaningful learning analytics tools.

Study boundaries and limitations

This study was conducted within the context of three Swedish municipalities that vary in size, data infrastructures, and use of DLMs. While this multi-context implementation provided valuable insights into scalability and adaptation, it also limits the generalizability of the findings to other educational systems or countries with different governance structures and data management practices.

The sample included 25 teachers, of whom 18 completed the post-implementation survey. Although this number provided rich qualitative insights, it represents a relatively small population, which constrains the ability to generalize findings statistically. Teacher experiences and perceptions reported in the study therefore reflect localized practices and should be interpreted as context-specific. The study relied primarily on self-reported data from interviews, surveys, and usability testing rather than continuous usage logs. This choice aligned with the ethical framework of the project but limited opportunities to analyze real-time dashboard interaction patterns. Consequently, the results focus on perceived usability and pedagogical value rather than measured behavioral outcomes. Furthermore, student performance data were used only in aggregated and anonymized form to protect privacy; no direct measurement of learning impact was conducted. The results therefore pertain to the design and perceived utility of the MLADs rather than to demonstrable changes in learning outcomes.

Overall, these boundaries reflect the exploratory nature of this DBR. The findings should be viewed as design insights and transferable principles rather than generalizable causal claims. Future studies could extend this work by including larger samples, diverse educational contexts, and longitudinal usage data to further validate and expand the results.

Conclusion and future work

While this study provides valuable insights into the potential of MLADs for supporting data-driven decision-making in primary education, it also highlights key challenges related to data presentation, teacher training, and scalability. By addressing these challenges and building on the promising findings of this research, future work can further enhance the effectiveness and impact of MLADs, ultimately leading to better outcomes for both teachers and students. For instance, future iterations could include recommendation systems that suggest instructional strategies based on student data, further helping teachers to translate data into actionable practices.

Another area of future work lies in exploring the scalability of the MLADs across different educational contexts. As the results of this study indicated, the variations in teaching practices, data sources, and technological infrastructures across municipalities present a

significant challenge for the widespread implementation of a single dashboard solution. To address this, future research should focus on developing adaptable systems that can be easily customized to meet the specific needs of different schools, districts, or regions. One promising direction is the development of modular dashboards, where components can be selected or tailored based on local requirements. This approach could allow the dashboards to be more flexible and responsive to the needs of diverse educational contexts.

Furthermore, while this study focused on the usability and functionality of the MLADs, the long-term impact of such tools on teaching practices and student outcomes remains an open question. In future work, we will include longitudinal studies to examine the sustained effects of the developed MLADs on both teacher decision-making and student learning outcomes. It would be valuable to assess how teachers' practices evolve over time as they become more familiar with using data-informed tools and whether these changes result in measurable improvements in student performance. Additionally, involving other stakeholders, such as students and parents, in the feedback process could provide a more holistic view of the dashboards' effectiveness and impact.

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Declarations

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